LAB#01

Fundamentals of Web Development & Web Designing

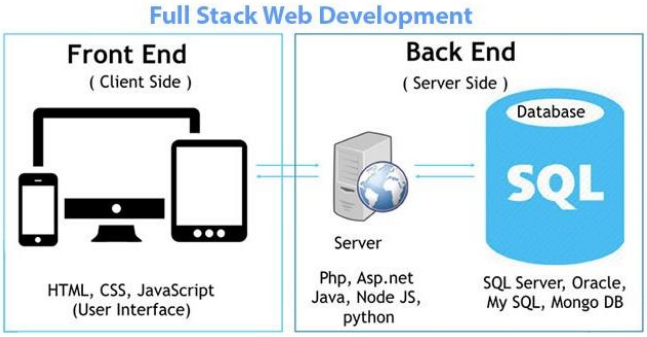
OBJECTIVE: Learn and Understand Web Development & Web Designing

- Introduction to web development along with web development concepts using HTML.
- Explore advanced HTML tags and concepts for enhanced functionality and interactivity in web development.
- To design webpages using HTML and CSS.

THEORY:

Introduction to Web Development

It is the process of building, creating and maintaining of websites. It involves coding and using programming languages to create websites that are functional, user-friendly, and perform well.



It includes aspects such as;

- Web designing
- Web Programming
- Web Publishing
- Database Management

It is the creation of an application that work over the internet. The main components of web development can be categorized as;

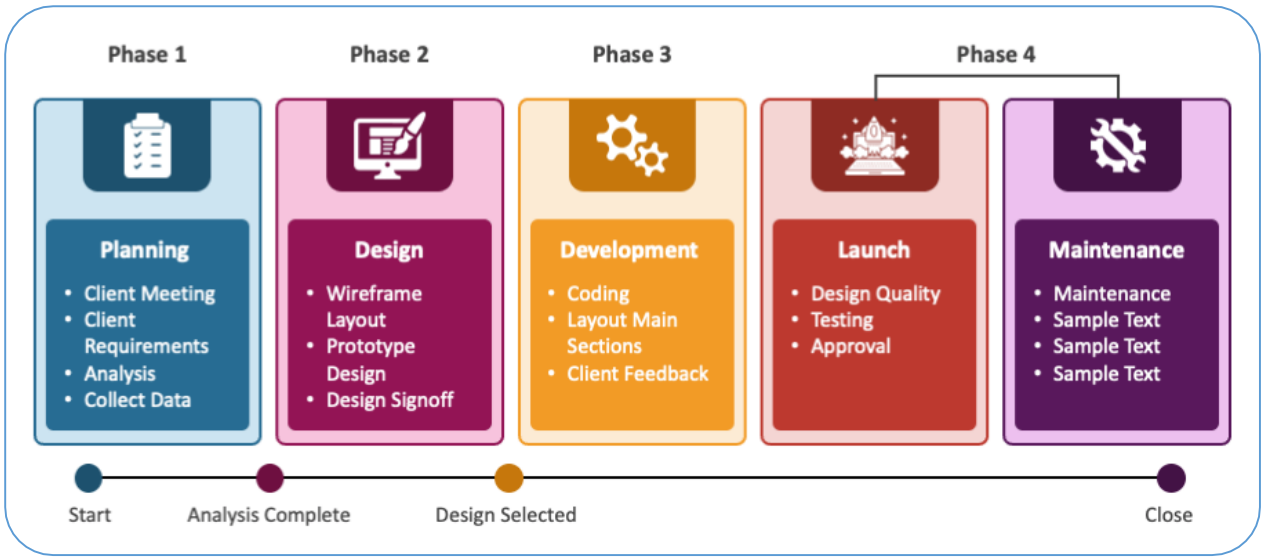
- **Front-end:** The part of the website that users see and interact with (HTML, CSS, JavaScript) also known as client-side development.

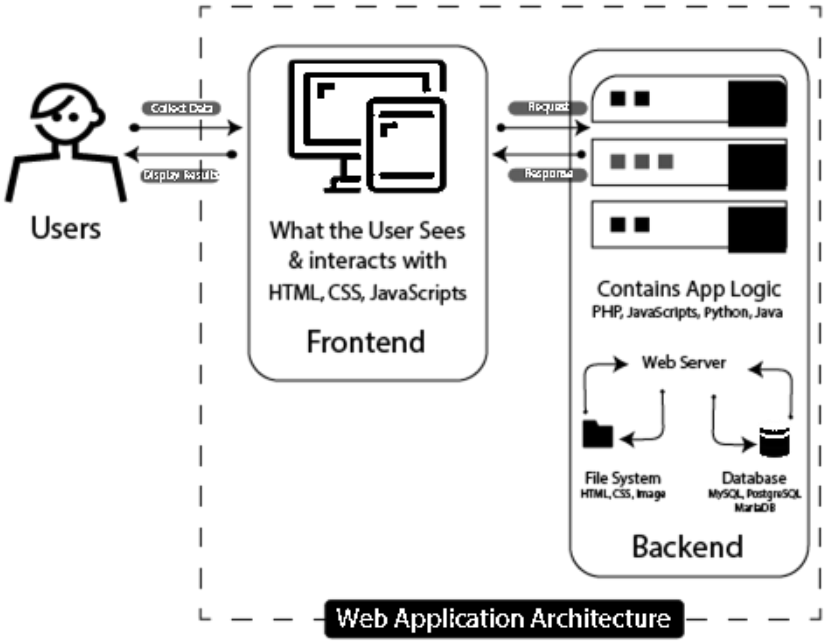
- **Back-end:** Handles server-side logic, databases, and data processing that users don't see directly. Common languages are Python, Java, Node.js, etc.
- **Full stack development:** Involves both frontend and backend work. A full-stack developer builds both the visible & the server parts of a website or app.

Web Development Workflow

The web development process involves several steps to ensure a systematic approach to building websites. The key stages are:

- **Planning**: Define the purpose of the website, target audience, and desired features.
- **Designing**: Create wireframes and mockups to visualize the layout and functionality.
- **Development**: Implement the design using HTML, CSS, and JavaScript for the frontend and backend technologies for server-side functionality.
- **Testing**: Check the website for bugs, responsiveness, and compatibility across devices and browsers.
- **Deployment**: Host the website on a web server and make it accessible over the internet.
- **Maintenance**: Update and maintain the website to ensure its functionality and relevance.



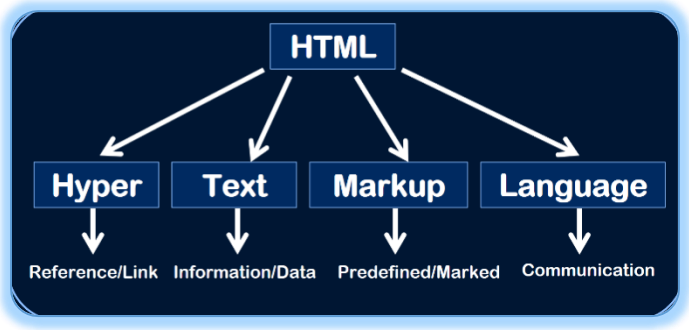


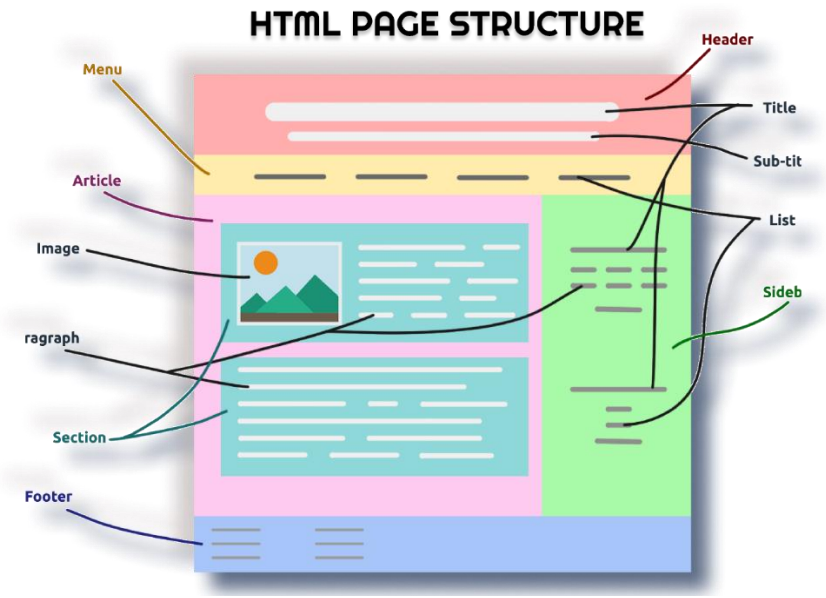
Introduction to Static & Dynamic Website

- **Static Website:** Contains fixed content that doesn't change without modifying the code. Good for simple sites with limited interactivity.
- **Dynamic Website:** Content can change in response to user interactions or server-side data. Requires backend logic and database support.

Introduction to HTML

HTML (HyperText Markup Language) is the standard language for creating web pages. It defines the structure of a webpage using tags that organize content into headings, paragraphs, links, images, etc. HTML elements tell the browser how to display the content.





Example Code

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
```

Example Explained

- The **<!DOCTYPE html>** declaration defines that this document is an HTML5 document
- The **<html>** element is the root element of an HTML page
- The **<head>** element contains meta information about the HTML page
- The **<title>** element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The **<body>** element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The **<h1>** element defines a large heading
- The **<p>** element defines a paragraph

HTML Tags and Defined Structure

Start tag	Element content	End tag
<h1>	My First Heading	</h1>
	My first paragraph.	
 	none	*none*

HTML Tags

Tags	***Purpose***	***Syntax***
Title Tag	It is used to specify the title of a web page, which is typically displayed in the browser's title bar.	**<title>** Page Title **</title>**
Heading Tags	It is used to display headings **h1, h2, h3, h4, h5, h6**	**<h1>** Hello World **<h1/>**
Paragraph Tag	It is used to add paragraphs.	** ** A Paragraph** **
Br Tag	It is used to add next line (line breaks) to our page.	** **
Bold, Italic & Underline Tags	These Tags are used to highlight text to our page.	****SSUET**** ****SSUET**** **<u>**SSUET**</u>**
Center Tag	It is used for centering content within a web page.	** ** Content** **
Subscript &Superscript Tag	They are used to display a horizontal ruler, used to separate content.	The**_{**subscripted**}** The**^{**superscripted**}**
Button Tag	It is used to create a clickable button in HTML. It can be used to perform certain actions i.e. submit, reset etc.	**<button>**Click Me!**</button>**

HTML Attributes and Defined Structure

Attributes are used to add more information to the tag. All HTML elements can have attributes. Some Attributes are **bgcolor href, src, width, height** etc.

Image Tag	Image Tag is used to add images to our Html page.	**<img src="link" height=50px width=50px >**
Iframe Tag	It is used to embed content from other websites i.e. YouTube, a Google Map etc.	**<iframe** src = "URL"width = "pixels" height = "pixels"**> </iframe>**
Video Tag	It is used to add video	**<video** src="myVid.mp4"**>**My Video **</video>**
Anchor Tag	Anchor Tag is used to add Links to our Html page.	**<a** href="https://google.com"**>** Google **</a>**

HTML Lists

Lists are used to represent real life list data.

Example Output

An unordered HTML list:

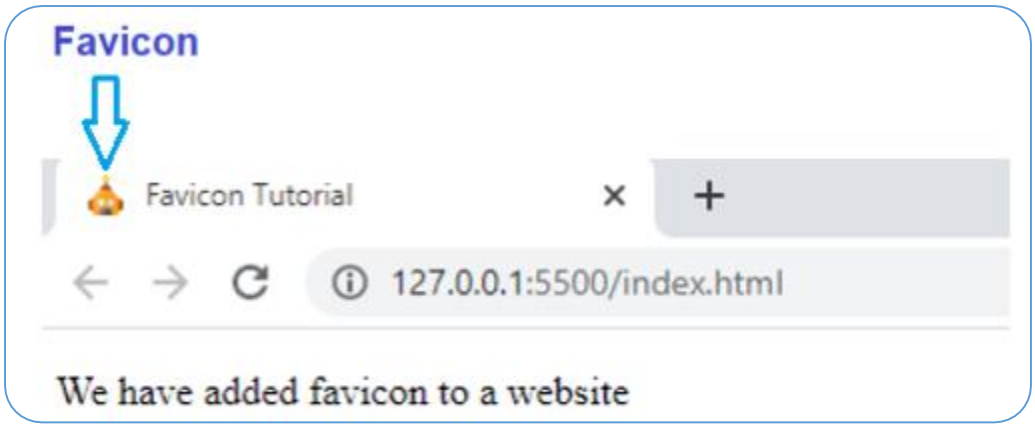
- Item
- Item
- Item
- Item

An ordered HTML list:

1. First item
2. Second item
3. Third item
4. Fourth item

HTML Favicon

A favicon is a small image displayed next to the page title in the browser tab.



To add a favicon to your website, either save your favicon image to the root directory of your webserver, or create a folder in the root directory called images, and save your favicon image in this folder. A common name for a favicon image is **"favicon.ico".**

Sample Code

```
<!DOCTYPE html>
<html>
<head>
  <title>My Page Title</title>
  <link rel="icon" type="image/x-icon" href="/images/favicon.ico">
</head>
<body>

<h1>This is a Heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

Favicon File Format Support

The following table shows the file format support for a favicon image:

Browser	ICO	PNG	GIF	JPEG	SVG
Edge	Yes	Yes	Yes	Yes	Yes
Chrome	Yes	Yes	Yes	Yes	Yes
Firefox	Yes	Yes	Yes	Yes	Yes
Opera	Yes	Yes	Yes	Yes	Yes
Safari	Yes	Yes	Yes	Yes	Yes

Labels and Input Handling

- The HTML <input> element is the most used form element.
- An <input> element can be displayed in many ways, depending on the type attribute.

Type	Description
<input type="text">	Displays a single-line text input field
<input type="radio">	Displays a radio button (for selecting one of many choices)
<input type="checkbox">	Displays a checkbox (for selecting zero or more of many choices)
<input type="submit">	Displays a submit button (for submitting the form)
<input type="button">	Displays a clickable button

<label> element

It is used to associate text labels with form elements, such as input fields, checkboxes, or radio buttons. Labels improve accessibility and usability by allowing users to click on the label to activate the corresponding input element.

Syntax

The **<label>** tag can be used in two ways:

1. Using the for Attribute

The **for** attribute connects the **<label>** to a specific input element via the id of the input.

Sample Code

```
<label for="username">Username:</label>
<input type="text" id="username" name="username">
```

2. Wrapping the Input Element

The input element is nested inside the **<label>,** eliminating the need for the for attribute.

Sample Code

```
<label>
   Password:
   <input type="password" name="password">
</label>
```

Examples of Labels

1. Text Input Code

```
<label for="email">Email Address:</label>
<input type="email" id="email" name="email">
```

2. Radio Buttons Code

```
<label>
   <input type="radio" name="gender" value="male"> Male
</label>
<label>
   <input type="radio" name="gender" value="female"> Female
</label>
```

3. Checkboxes Code

```
<label for="subscribe">
   <input type="checkbox" id="subscribe" > Subscribe to newsletter
</label>
```

4. Dropdown Menu Code

```
<label for="country">Select Country:</label>
<select id="country" name="country">
   <option value="usa">USA</option>
   <option value="canada">Canada</option>
   <option value="uk">UK</option>
</select>
```

Forms along with Labels and Input Handling

- Use **forms** to collect data from users.
- Introduce various input types (text, password, email, number, date, checkbox, radio, file, etc.).

Sample Code

```
<form>
   <label for="username">Username:</label>
   <input type="text" id="username" name="username" required>
   <label for="email">Email:</label>
   <input type="email" id="email" name="email" required>
   <button type="submit">Submit</button>
</form>
```

HTML div Tag

It is used to create divisions on a web page. It is used to group together related elements, like paragraphs, images, and links.

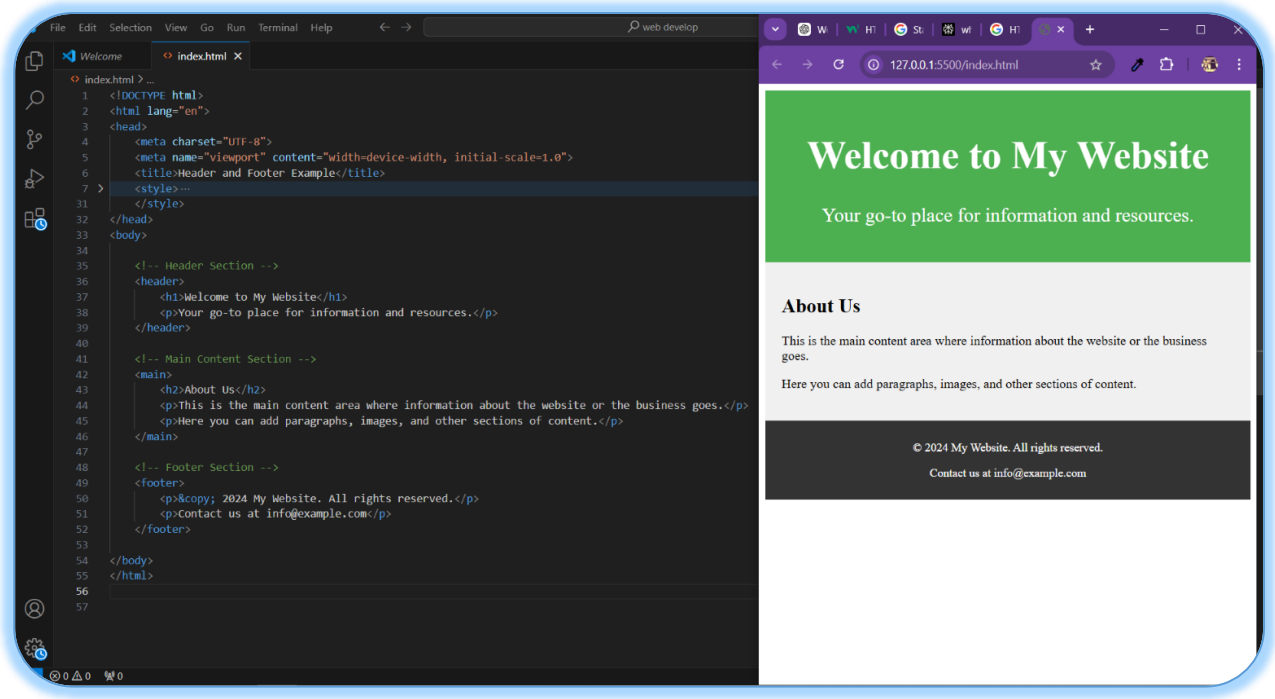
**Syntax: ……. **

```html
<!DOCTYPE html>
<html>
<head>
<style>
.myDiv {
 border: 3px outset black;
 background-color: lightblue;
 text-align: center;
}
</style>
</head>
<body>
<h1>The div tag</h1>
<div class="myDiv">
 <h2>Heading in a div tag</h2>
 <p>Text in a div tag.</p>
</div>

<p>Text outside the div tag.</p>

</body>
</html>
```

HTML header, footer and main Tags

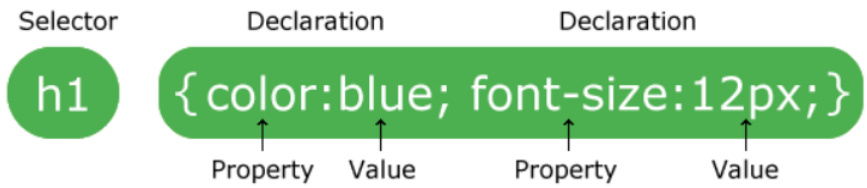


Introduction to CSS

CSS (Cascading Style Sheets) is a stylesheet language used to describe the presentation of a webpage. CSS styles HTML elements by controlling aspects like colors, fonts, layouts, and spacing.

CSS Syntax

A CSS rule consists of a ***selector*** and a ***declaration*** block.



1) The **selector** points to the HTML element you want to style.
2) The declaration block contains one or more declarations separated by **semicolons**.
3) Each declaration includes a CSS property **name** and a **value**, separated by a colon.
4) A CSS declaration always ends with a **semicolon**, and declaration blocks are surrounded by **curly braces**.

```html
<!DOCTYPE html>
<html>
<head>
<style>
p {
  color: red;
  text-align: center;
}
</style>
</head>
<body>

<p>Hello World!</p>
<p>These paragraphs are styled
with CSS.</p>

</body>
</html>
```

Hello World!

These paragraphs are styled with CSS.

How to Use CSS?

CSS can be applied in three main ways:

1) **Inline CSS:** Adds styles directly within HTML tags using the style attribute.

```
<h1 style="color: blue;">Hello, World!</h1>
```

2) **Internal CSS:** Uses a <style> tag inside the <head> section to define styles for the entire page.

```
<head>
  <style>
    h1 { color: blue; }
    p { font-size: 16px; }
  </style>
</head>
```

3) **External CSS:** Links to an external .css file, ideal for keeping HTML and CSS separate.

```
<head>
  <link rel="stylesheet" href="styles.css">
</head>
```

Introduction to Class & ID in HTML?

In HTML, **class** and **id** are attributes used to assign styles or behaviors to specific elements. They help in targeting elements in **CSS** and **JavaScript** for styling or scripting purposes.

Class: It is a reusable attribute that can be applied to multiple elements on a page. It can be used when we want the same styling or behavior across different elements.

Syntax in HTML: class="classname"

```html
<!DOCTYPE html>
<html>
<head>
    <style>
        /* CSS to style elements with the "highlight" class */
        /* In CSS, Classes are denoted with a dot (.) before the class name. */
        .highlight {
            color: rgb(37, 15, 234);
            background-color: rgb(232, 225, 85);
            padding: 5px;
        }
    </style>
</head>
<body>

    <p class="highlight">This paragraph is highlighted.</p>
    <p>This paragraph is not highlighted.</p>
    <p class="highlight">Another highlighted paragraph.</p>

</body>
</html>
```

This paragraph is highlighted.

This paragraph is not highlighted.

Another highlighted paragraph.

ID: It is a unique identifier for an element. Each id should be unique on the page, meaning it should be used only once per element.

Syntax: id="uniqueID"

```html
<!DOCTYPE html>
<html>
<head>
    <style>
        /* CSS to style the element with the "SSUET" id */
        /* In CSS, IDs are denoted with a hash (#) before the id name. */
        #SSUET {
            color: white;
            background-color: green;
            padding: 10px;
            text-align: center;
        }
    </style>
</head>
<body>

    <div id="SSUET">This is the header section.</div>
    <p>This is a regular paragraph.</p>

</body>
</html>
```

This is the header section.

This is a regular paragraph.

Common CSS Properties

Name	Purpose	Syntax
Color	Sets the color of text	p { color: red; }

Background-color	Sets the background color of an element.	body {
		background-color: lightgray;
		}

Font-family	Changes the font style	h1 {
		font-family: Arial, sans-serif;
		}

Font-size	Sets the size of the text	p { font-size: 18px; }

Text-align	Aligns text within an element (e.g., left, center, right).	h2 {
		text-align: center;
		}

Margin	Adds space outside an element's border.	.box {
		margin: 20px;
		}

Padding	Adds space inside an element's border.	.content {
		padding: 15px;
		}

Student Task:

Create a multi-page website using HTML and CSS that includes a **Login Page**, **Registration Page**, and **Home Page** with a **Sidebar**. Follow the instructions below:

Pages to Create:

1. **Login Page:**
    - Create a form with the following fields:
        - Username (text input)
        - Password (password input)
    - Add a "Login" button.
2. **Registration Page:**
    - Create a form with the following fields:
        - Username (text input)
        - Email (email input)
        - Password (password input)
        - Confirm Password (password input)
        - Gender Field
    - Add a "Register" button.
3. **Home Page:**
    - Add a heading like "Welcome to [Your Site Name]".
    - Add a **Sidebar** on the left side of the page with the following links:
        - Home
        - Login
        - Registration

Requirements:

- Use **HTML** for the structure and include the <header>, <footer>, and <main> tags where applicable.
- Use the <form> tag for user input and include proper <label> elements for accessibility.
- The **Sidebar** should be inside a and use the <ul> and <li> elements to list links.
- Ensure the pages are interconnected with proper hyperlinks.
- Use a favicon for your website.
- Add basic validation for required fields using the required attribute.

Hands on Practice Task (Optional)

Create a personal portfolio website using **HTML & CSS**. Your portfolio should include the following sections:

1. **Header:** Include your name and a short tagline or description about yourself.
2. **About Me Section:** Write a brief paragraph introducing yourself, your background, or your interests.
3. **Projects Section:** List at least 3 of your recent or favorite projects (you can create placeholders if you don't have real projects).
4. **Contact Section:** Provide contact details, such as your email address, or a simple message like "Get in touch!"

Requirements:

- Use semantic HTML tags (e.g., <header>, <section>, <footer>).
- Ensure proper use of headings (<h1> to <h3>).
- Use lists (<ul> or <ol>) where appropriate.
- Add at least one hyperlink (e.g., a link to a project or a placeholder email link).